

College Placement Tracker (MERN Stack)

A Complete Development Guide

This document provides a step-by-step guide to build a College Placement Tracker using the MERN stack (MongoDB, Express, React, Node.js).

1. Project Overview

The Placement Tracker helps students track job applications and allows admins to manage placement data efficiently.

2. Tech Stack

Frontend: React (Vite) + Tailwind CSS

Backend: Node.js + Express

Database: MongoDB (Mongoose)

Authentication: JWT

3. Backend Setup

Step 1: Initialize project using npm init

Step 2: Install dependencies: express, mongoose, cors, dotenv, jsonwebtoken

Step 3: Create server.js and connect MongoDB

Step 4: Setup folder structure (models, routes, controllers)

4. API Development

Auth APIs: Register and Login using JWT

Company APIs: Add, update, delete, and fetch companies

Application APIs: Apply to companies and update status

5. Frontend Development

Create React app using Vite

Setup routing using react-router-dom

Create pages: Login, Dashboard, Companies, Applications, Admin Panel

Use Axios for API calls and store token in localStorage

6. Key Features

Students can apply to jobs and track status

Admins can manage companies and update results

Dashboard shows statistics

7. Deployment

Deploy backend using Render

Deploy frontend using Vercel

Connect frontend with backend API

8. Enhancements

Add charts for analytics

Add email notifications

Add real-time updates using Socket.io

Conclusion

This project demonstrates full-stack development skills and is suitable for portfolios and interviews.